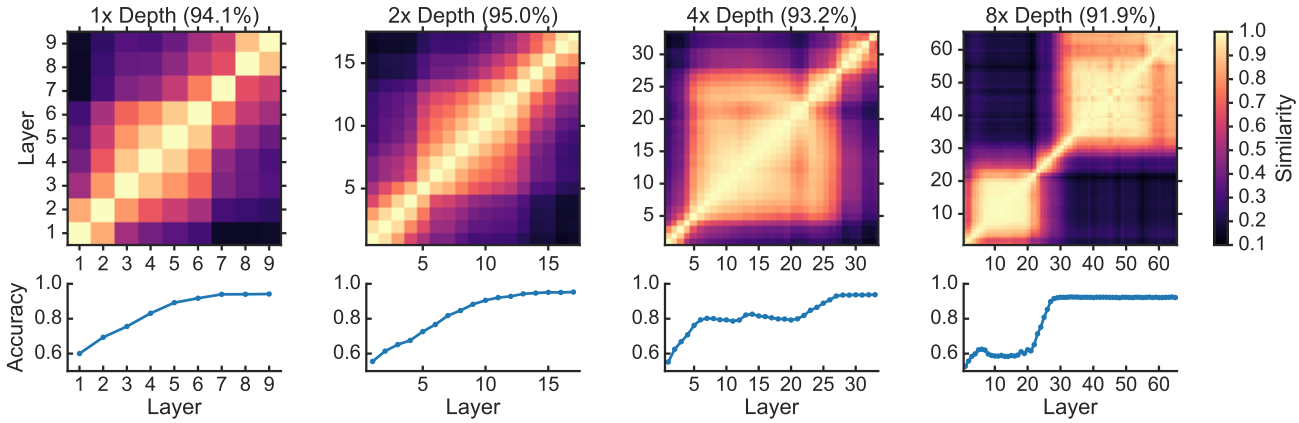

Figure 3. CKA reveals when depth becomes pathological. **Top**: Linear CKA between layers of individual networks of different depths on the CIFAR-10 test set. Titles show accuracy of each network. Later layers of the 8x depth network are similar to the last layer. **Bottom**: Accuracy of a logistic regression classifier trained on layers of the same networks is consistent with CKA.

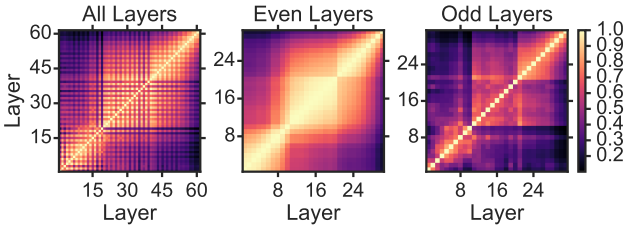

Figure 4. Linear CKA between layers of a ResNet-62 model on the CIFAR-10 test set. The grid pattern in the left panel arises from the architecture. Right panels show similarity separately for even layer (post-residual) and odd layer (block interior) activations. Layers in the same block group (*i.e.* at the same feature map scale) are more similar than layers in different block groups.

network, accuracy plateaus less than halfway through the network. When applied to ResNets (He et al., 2016), CKA reveals no pathology (Figure 4). We instead observe a grid pattern that originates from the architecture: Post-residual activations are similar to other post-residual activations, but activations within blocks are not.

CKA is equally effective at revealing relationships between layers of different architectures. Figure 5 shows the relationship between different layers of networks with and without residual connections. CKA indicates that, as networks are made deeper, the new layers are effectively inserted in between the old layers. Other similarity indexes fail to reveal meaningful relationships between different architectures, as we show in Appendix F.5.

In Figure 6, we show CKA between networks with different layer widths. Like Morcos et al. (2018), we find that increasing layer width leads to more similar representations between networks. As width increases, CKA approaches 1; CKA of earlier layers saturates faster than later layers. Networks are generally more similar to other networks of the same width than they are to the widest network we trained.

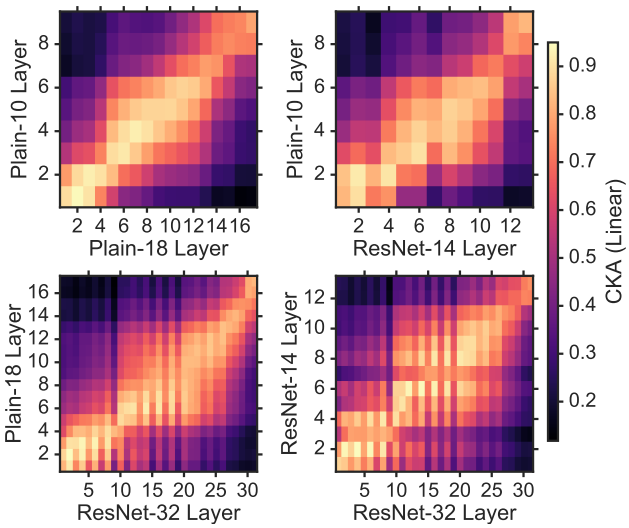

Figure 5. Linear CKA between layers of networks with different architectures on the CIFAR-10 test set.

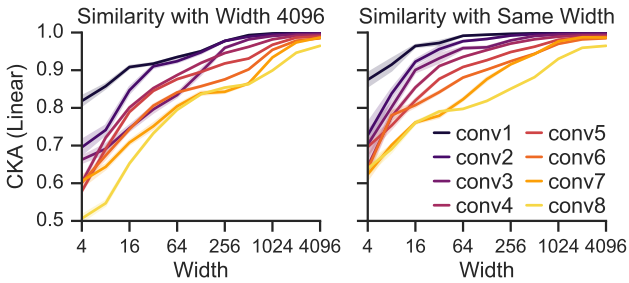

Figure 6. Layers become more similar to each other and to wide networks as width increases, but similarity of earlier layers saturates first. **Left**: Similarity of networks with the widest network we trained. **Middle**: Similarity of networks with other networks of the same width trained from random initialization. All CKA values are computed between 10 networks on the CIFAR-10 test set; shaded regions reflect jackknife standard error.